

Chapter 7 — Discussion

7.1 Introduction

The empirical chapters of this dissertation produced four findings whose theoretical and methodological implications now require integration. First, the contemporary European education–fertility gradient is non-monotonic: medium-educated women record the lowest fertility, with high-educated women recording fertility above medium but below low across most of the panel. Second, cross-country variation in gradient steepness cannot be definitively attributed to either cultural or economic country-level characteristics at $N = 21$, though the economic block holds a narrow inferential edge. Third, the individual-level effect of education on women's completed fertility in the BCS70 cohort is mediated primarily by realised earnings, not by attitudinal pathways — contradicting the dissertation's own H4 hypothesis and constraining the cultural-transmission account of fertility decline at the individual level. Fourth, the U-shape signature, while robust across countries, is not robust to population weighting, indicating that it characterises European countries as units of analysis more than European women as a population. This chapter situates these findings in the literature reviewed in Chapter 2, discusses their theoretical implications, acknowledges the limitations of the empirical strategy, and identifies directions for future research.

7.2 Summary of Findings

H1 holds at the European population level. The two-way fixed-effects estimate of the high-versus-low contrast in log eTFR is -0.187 ($p = 0.002$), corresponding to approximately 17% lower fertility among high-educated women than among low-educated women across the pooled 21-country panel. This finding survives four of the five robustness specifications

considered, including the balanced subsample, the exclusion of pandemic years, the exclusion of countries with composition-driven small low-education tiers, and the alternative reference coding. Under population weighting the negative low–high gradient remains robust ($\beta = +0.242$ on the low contrast versus the medium reference, $p = 0.006$).

H2 is confirmed parametrically and distributionally. The quadratic specification produces a positive and precisely estimated curvature coefficient ($\beta_2 = +0.195$, $p < 0.001$), and the AIC and BIC differences against the linear specification ($\Delta\text{AIC} = -204$, $\Delta\text{BIC} = -199$) both decisively favour the quadratic functional form. The curve minimum falls at education = 1.24 on the ordinal scale (low = 0, medium = 1, high = 2), placing the lowest predicted fertility between the medium and high tiers and closer to medium. Quantile regressions at five quantiles confirm the U-shape across the entire conditional distribution of log eTFR, though the U-shape's left and right arms exhibit different distributional sensitivities.

H3 is unsupported in the strong sense that no model produces a definitive ranking of cultural over economic predictors. The XGBoost SHAP decomposition assigns 59.5% of in-sample importance to cultural indicators combined (institutional plus attitudinal) versus 40.5% to economic indicators; the Bayesian model comparison reverses this, ranking the economic model best on both WAIC and LOO criteria. The LOOCV R^2 of -0.086 indicates that macro-level indicators lack reliable out-of-sample predictive power at this sample size. The most precisely identified single parameter across all six Bayesian models is the `gdp_percap` coefficient at posterior median -0.27 with $P(< 0) = 0.97$ — economic, not cultural.

H4 is contradicted by the data. The DML mediation decomposition shows that for women, the economic pathway through earnings accounts for 65.6% of the total causal effect of education on completed fertility, while the attitudinal pathway accounts for only 3.8%. The

two pathways are an order of magnitude apart, and the qualitative ordering is the opposite of what H4 predicted. For men, the total effect is small and statistically marginal, and the pathway decomposition shows suppression rather than mediation, with the direct effect exceeding the total effect in absolute magnitude.

7.3 Theoretical Implications

Three theoretical contributions follow from these findings. The first is descriptive but substantively important: the contemporary European education–fertility gradient is dominantly U-shaped, with medium-educated women at the bottom of the fertility distribution in twelve of twenty-one countries in the panel. The classical demographic-transition framing — in which fertility falls monotonically with socioeconomic resources — is too simple. The Stulp and Barrett (2016) review identified the J-curve hypothesis as theoretically important but empirically fragmentary; the systematic cross-country test conducted here with education-stratified period fertility data, formal distributional methods, and the gradient-shape classification documented in Chapter 4 fills that gap. The headline contribution is not merely the detection of non-monotonicity in some countries but the demonstration that the U-shape is the dominant pattern, not the exception.

The second contribution concerns the cultural-transmission account of fertility decline. The cultural-transmission framework, as developed by Bongaarts and Watkins (1996) and Colleran (2016, 2020), makes a joint prediction at two levels of analysis: that fertility norms propagate through populations via social transmission of low-fertility attitudes from high-status individuals to lower-status ones, and that the proximate mechanism through which education depresses individual fertility is the change in attitudes that education induces. The dissertation's H3 and H4 results separately constrain this joint prediction. At the individual level (H4), the data directly contradict the proximate-mechanism claim: attitudes account for less than 4% of the effect of education on women's completed fertility in the

BCS70 cohort, while earnings account for 66%. At the country level (H3), the cultural block does not dominate the economic block on inferential criteria, and the most precisely identified single predictor is GDP per capita. Both findings, read together, place the cultural-transmission account under more pressure than either does alone.

The macro and micro findings, however, do not unambiguously refute cultural-transmission theory. A more nuanced reading is available. Cultural-transmission theory may still be correct about how fertility norms propagate across populations and across time, while being wrong about how education depresses fertility within individuals at a given moment. These are different theoretical claims operating on different timescales. The propagation of low-fertility norms through social networks is a long-run population-level phenomenon; the proximate mechanism through which individual education translates into individual fertility outcomes is a short-run individual-level phenomenon. The dissertation's evidence speaks more directly to the latter than to the former. A cultural-transmission theorist could accept the H4 finding (that earnings dominate individual mediation) while maintaining that the broader cross-cohort decline in European fertility — the phenomenon the framework was originally developed to explain — propagates through cultural transmission. Disentangling these levels theoretically is beyond the scope of this dissertation but is a productive direction for the literature.

The third contribution concerns the demographic transition fitness paradox itself. Vining (1986) framed the paradox as a uniform negative gradient that required explanation in evolutionary terms; the J-curve literature has subsequently suggested the pattern is more limited and that fertility may revive at the upper end of the SES distribution (Hopcroft, 2006, 2015; Fieder and Huber, 2007). The dissertation's findings refine the paradox further. The negative gradient is real at the low–high comparison (H1), but the bottom of the gradient is located at medium education rather than at high education (H2), meaning that the most highly

educated women do not have the lowest fertility. Under the quantity–quality framework, this is what one would expect if high-educated women face the lowest absolute opportunity cost relative to their earning power — they can outsource childcare, maintain career trajectories, and convert high incomes into fertility-supporting investments more efficiently than medium-educated women, who face the steepest opportunity costs because they have meaningful wage penalties but insufficient earnings to neutralise them. This interpretation is speculative at the level of mechanism but is consistent with both the macro U-shape and the H4 finding that earnings dominate individual mediation.

Finally, the maladaptation framework introduced in Chapter 2 (Pérusse, 1993; Kanazawa, 2003) is not directly tested by the dissertation's analyses but is implicitly engaged by the H4 finding. The maladaptation account treats modern low fertility as a mismatch between ancestral psychology and modern environments — resources are diverted into status competition rather than reproduction not because doing so maximises current fitness but because the underlying psychology was shaped by ancestral conditions. The H4 finding that the dominant individual-level mechanism is realised opportunity cost rather than transmitted attitudes is more consistent with a rational-choice flavour (quantity–quality substitution) than with a mismatch flavour (evolved psychology systematically misfiring in modern environments). This does not refute the maladaptation framework — one could argue that responding to wage incentives by reducing fertility is itself an evolved mechanism that produces maladaptive outcomes in modern environments — but it does shift the burden of explanation toward accounts that have rational economic responses as their proximate mechanism rather than evolved psychological heuristics.

7.4 Comparison with Prior Literature

The H4 finding contradicts Snopkowski and Sear (2013), who used Indonesian data to argue that the attitudinal pathway was substantially more important than the economic

pathway in predicting fertility intentions among educated women. The disagreement is informative rather than problematic. Snopkowski and Sear's study used a Southeast Asian developing-country setting in which educational expansion was rapid and the labour-market opportunity costs of childbearing for educated women were modest in absolute terms; the BCS70 cohort is a British 1970 birth cohort in a high-income, post-industrial labour market where educated women's earnings are substantial in absolute terms and the opportunity cost of childbearing is therefore much larger. The directional difference between the two studies is consistent with the structural prediction that opportunity-cost mechanisms dominate where wages are high, while attitudinal mechanisms may dominate where wages are low. This is not a refutation of either finding; it is a substantive cross-context regularity that the literature should investigate systematically.

The H4 finding extends Berrington's (2017) work on the BCS70 cohort. Berrington documented divergence between fertility intentions and realised fertility for highly educated women in BCS70, attributing it primarily to delayed childbearing and unintended childlessness; she did not run a formal causal mediation analysis. The dissertation's DML mediation decomposition provides the causal-inference complement to her descriptive work: not only does extended education postpone childbearing in this cohort, but the postponement is mediated principally by the earnings consequences of education, not by the attitudinal changes education induces. This is a methodological extension rather than a substantive disagreement.

Goodman, Koupil and Lawson (2012) used multigenerational Swedish data to argue that low-fertility, high-investment strategies do not maximise long-run reproductive success measured in grandchildren or great-grandchildren. The dissertation's H4 finding speaks to a different question — the proximate mechanism within a generation rather than the multigenerational fitness consequences — but is methodologically consistent. Both studies

highlight the role of opportunity-cost mechanisms in producing the contemporary low-fertility gradient, and both findings are consistent with the broader picture in which education depresses fertility through economically rational responses to changed opportunity structures, even when those responses do not maximise biological fitness.

Hopcroft (2006, 2015) reported positive fertility-income gradients among married high-income men in US General Social Survey data, interpreting this as evidence that the demographic transition fitness paradox is partly an artefact of pooling married and unmarried individuals. The dissertation's H4 estimates were not stratified by marital status, but the partnership pathway in the mediation decomposition (positive +0.032 for women, recovering 22.7% of the total effect) is broadly consistent with Hopcroft's interpretation: partnered status is positively associated with fertility, highly educated women are more likely to be partnered at age 42, and the partnership pathway therefore offsets some of the negative earnings pathway. A formal extension of the H4 analysis stratifying by partnership history is a direction for future work.

More broadly, the H3 cross-country analysis provides the systematic multi-country test of the J-curve hypothesis that the Stulp and Barrett (2016) review identified as missing from the literature. Their conclusion that the SES-fertility relationship is heterogeneous across populations, sub-groups, and historical periods is confirmed and refined by the dissertation's gradient-shape classification: the U-shape is the dominant pattern but coexists with five monotonic-negative and two inverted-bottom countries, and the cross-country heterogeneity is not explained by any single cultural or economic predictor at $N = 21$.

7.5 Strengths of the Empirical Strategy

Three methodological features of the dissertation are worth noting. First, the H4 analysis applies the Double Machine Learning framework of Chernozhukov et al. (2018) to the BCS70 cohort. The framework's combination of random-forest nuisance estimation,

sample-splitting cross-fitting, and Neyman-orthogonal moment conditions is appropriate to the high-dimensional confounding problem inherent in observational mediation analysis. To my knowledge, this is the first application of DML mediation decomposition to the BCS70 cohort's age-30 to age-42 sweeps. The age-51 sweep released in 2023 has not previously been used in combination with the age-30 NVQ measure for causal mediation analysis of fertility outcomes.

Second, the H3 analysis combines XGBoost-based discovery with Bayesian inference in a deliberately staged way, and treats the disagreement between the two stages as a substantive finding rather than as a problem to be reconciled. The two-stage discovery–inference design (Athey and Wager, 2019; Lundberg and Lee, 2017, on the discovery side; Bürkner, 2017, on the inference side) is increasingly common in causal-inference and policy-evaluation literatures but remains rare in demographic applications. The dissertation's transparent reporting of the WAIC and SHAP rankings, together with the LOOCV R^2 result that frames the limits of macro-level inference at $N = 21$, demonstrates methodological honesty about what country-level data can and cannot establish about cross-country variation in the gradient.

Third, the H1 and H2 analyses combine two-way fixed-effects panel regression, quadratic specification, and quantile regression. The combination is more informative than any one of these methods alone: the fixed-effects model establishes the categorical comparison cleanly, the quadratic specification tests the U-shape parametrically, and the quantile regressions test whether the U-shape holds across the distribution of fertility. The five-specification robustness battery — balanced subsample, no-COVID, no-Type-A, alternative reference coding, and population weighting — exposes the U-shape's sensitivity to population weighting honestly rather than reporting only the specifications that produce the strongest result.

A more diffuse strength is the dissertation's organisation around hypothesis-level findings rather than around a single descriptive narrative. Each of the four hypotheses produced a distinct verdict (H1 supported, H2 supported with qualification, H3 unsupported in the strong sense, H4 contradicted), and the dissertation states each verdict cleanly. Honest reporting of mixed and falsifying findings is a contribution in its own right at a time when the broader literature on the demographic transition fitness paradox has accumulated a substantial number of contradictory results without a corresponding accumulation of resolution.

7.6 Limitations

Several limitations qualify the findings and warrant explicit acknowledgement. The most important is the pivot from completed cohort fertility, as originally proposed, to period education-stratified TFR computed from Eurostat data. The Wittgenstein Centre's R interface exposed only projection data rather than the historical reconstruction series that the proposal assumed would be accessible, forcing the substitution. Period eTFR is a synthetic-cohort measure that conflates the timing and quantum of fertility; the U-shape pattern observed could in principle reflect differential postponement across educational groups rather than differential completed fertility. The 2024 cut-off of the panel mitigates this concern for older cohorts within the panel but does not eliminate it. The substantive findings should be interpreted as describing the European period fertility regime over 2007–2024 rather than the completed lifetime fertility of any specific birth cohort.

The country sample of $N = 21$ is the binding constraint on inferential power for H3. The Bayesian common-case subset of $N = 18$ is smaller still. At these sample sizes, no model comparison can produce coefficient estimates whose 95% credible intervals cleanly exclude zero, and the LOOCV R^2 of -0.086 indicates that the XGBoost feature importances do not generalise out of sample. The H3 verdict — that neither cultural nor economic indicators clearly dominate — is therefore conditional on the sample size as well as on the indicators

chosen. A larger country sample with comparable education-stratified period fertility data, ideally combined with comparable measures of childcare costs and housing affordability, would be the most direct way to strengthen the macro-level evidence.

Several economic predictors that the literature has identified as substantively important are not represented in the H3 analysis. Childcare costs and coverage, which the OECD Family Database covers from 2015 onwards with substantial pre-2015 missingness, would arguably be the single most direct economic measure of the cost of children but could not be included on a consistent panel basis. Housing affordability, which the recent fertility literature has linked to postponement (Aassve, Mazzuco and Mencarini), is similarly absent. Welfare state generosity, family benefit expenditure, youth unemployment, and income inequality (Gini) would all have strengthened the economic block. Their omission means that the H3 economic-versus-cultural comparison is conducted with a narrower operationalisation of the economic block than the substantive literature would justify, and the modest WAIC advantage of the economic model may understate the true economic contribution to cross-country variation. This is a real constraint.

At the individual level (H4), the analysis is restricted to a single British 1970 birth cohort. The findings generalise with confidence only to broadly similar cohorts in broadly similar institutional and labour-market settings. The mediation result that earnings dominate the attitudinal pathway may not hold in countries where the wage returns to education are lower in absolute terms, where childcare and family benefits are more generous (reducing the earnings cost of childbearing), or where prevailing fertility norms are more variable across educational groups. The Snopkowski and Sear (2013) Indonesian finding cited in Section 7.4 is precisely such a contrary case.

Two timing limitations affect the H4 mediation decomposition. The attitudinal mediators (`m_trad_gender`, `m_profamily`) are measured at age 30, the same sweep from which NVQ

qualifications are drawn; they are therefore contemporaneous with the treatment rather than temporally downstream of it. The sequential ignorability assumption (Imai, Keele, and Tingley, 2010) is harder to defend when mediator and treatment are contemporaneous, because unmeasured common causes of both are more difficult to rule out. The economic mediator (log earnings at age 42) and partnership mediator (cohabiting partner at age 42) are measured at the same sweep as the outcome and so control for them as additional confounders of the education–fertility relationship rather than as true downstream mediators in the strict causal sense. Both limitations were anticipated in the analytical strategy (Section 3.8.2) but should be borne in mind when interpreting the pathway-specific estimates.

The U-shape finding from H2 is not robust to population weighting. Under the population-weighted M1 specification reported in Section 5.4, the medium-versus-high coefficient falls to -0.012 ($p = 0.890$), meaning that in larger countries the high-educated and medium-educated tiers have indistinguishable mean fertility once country and year effects are controlled. The U-shape is therefore a feature of European countries as units of analysis more than of European women as a population. This qualification, which has been stated openly in Chapters 5 and 6, narrows but does not eliminate the H2 finding: the country-level U-shape is real and consequential for cross-national demographic patterns, even if it does not characterise the European population at the individual level.

Several smaller limitations should also be noted. The wild-cluster bootstrap recommended for inference with only 21 clusters (Cameron, Gelbach and Miller, 2008) could not be implemented because the `fwildclusterboot` package is unavailable for R 4.5.1; the cluster-robust standard errors used for H1 inference are therefore subject to the standard caveat that asymptotic approximations may be imprecise at small cluster counts. The quantile regressions for H2 use the `quantreg` package, which does not support cluster-robust inference; standard errors for the quantile coefficients are not reported. The 12% threshold used to

distinguish composition-driven from broad U-shape countries in the gradient classification (Section 4.5) is an empirical choice rather than a theoretically derived value; sensitivity to alternative thresholds is reported in Chapter 5 but the choice itself remains a discretionary one. North Macedonia is absent from the European Social Survey, reducing the ESS country sample to 20; Romania contributes only one ESS round. The contraceptive prevalence indicator is missing for 9 of 21 countries, restricting its inclusion to the XGBoost stage only. None of these smaller issues alters the substantive conclusions, but each is a real constraint that an attentive reader is entitled to.

7.7 Future Research Directions

Five directions for future research follow directly from the dissertation's findings and limitations. The first is replication of the H4 analysis in other birth cohorts and other countries. The 1958 National Child Development Study (NCDS), accessible via the UK Data Service alongside BCS70, provides a 12-year-earlier British cohort with comparable structure; its inclusion would test whether the earnings dominance over attitudes is stable across British cohorts or specific to the 1970 cohort. The Wisconsin Longitudinal Study and the German Socio-Economic Panel offer comparable continental and US testing grounds. The directional comparison with the Snopkowski and Sear (2013) Indonesian finding could be sharpened by analyses in middle-income contexts where opportunity-cost mechanisms might be expected to be weaker.

The second is the extension of H3 to a larger country sample. The Wittgenstein Centre's historical reconstruction series, if accessible in machine-readable form, would extend the sample to thirty-five or more European and Anglo-Saxon countries on completed cohort fertility for cohorts born 1930–1980. With a country sample of fifty or more, the Bayesian comparison would no longer be constrained to credible intervals that cross zero, and the H3 question — whether cultural or economic predictors carry more weight in cross-country

variation — could be answered with substantially greater inferential power. Adding childcare cost and housing affordability indicators on a consistent panel basis, perhaps from 2015 onwards as the OECD Family Database expands its coverage, would also strengthen the economic block and produce a fairer cultural-versus-economic comparison.

The third is the use of partnership histories more carefully in H4. The BCS70 partnership history file (UKDA 6941, used in the partnership age-42 indicator but not in its full longitudinal richness in this dissertation) provides month-level partnership data from age 16 to age 51. A more careful treatment would model the timing and duration of partnerships rather than the snapshot at age 42, allowing the partnership pathway to be decomposed into selection (educated women being more likely to enter stable partnerships) and exposure (longer time spent in partnerships providing more reproductive opportunity). The dissertation's positive partnership-pathway estimate (+0.032 for women) hints that this is substantively important, but the underlying mechanism requires longitudinal modelling that the cross-sectional mediation decomposition cannot deliver.

The fourth is the engagement with partner-level economic variables. The H4 analysis used the BCS70 cohort member's own earnings as the economic mediator. Partner's earnings, where partner data are available in BCS70 sweeps from age 30 onwards, would allow the assortative-mating channel to be modelled directly: high-educated women may have lower fertility partly because they partner with high-educated, high-earning men whose own opportunity costs interact with the cohort member's. The partner-economic channel is theoretically important to the quantity–quality framework but is not separately identified in the dissertation's analysis.

The fifth direction is conceptually broader. The macro-versus-micro mechanism gap identified in Section 7.3 — the suggestion that cultural-transmission may operate as a population-level propagation mechanism while not being the dominant proximate

individual-level mediator — is a theoretical hypothesis that the dissertation cannot directly test. Investigating it would require a research design combining individual-level cohort data on attitudes and fertility with measurements of attitudinal diffusion across networks and across cohort boundaries, ideally with a comparative cross-country dimension. Such a design is methodologically demanding but would address the central unresolved question in the contemporary demographic transition literature: how individual-level rational economic responses and population-level cultural transmission jointly produce the demographic patterns we observe. Chapter 8 returns to this question briefly in its closing remarks.

7.8 Concluding Remarks

Taken together, the dissertation's findings sharpen rather than resolve the long-standing debate over the demographic transition fitness paradox. The empirical phenomenon (negative low–high gradient combined with a U-shape that places medium-educated women at the bottom of the fertility distribution) is more nuanced than the classical framing suggested. The individual-level mechanism (earnings-dominated mediation contradicting the attitudinal prediction of H4) constrains the cultural-transmission framework in the proximate sense while leaving open the possibility that cultural mechanisms operate at population-propagation scales. The country-level analysis at $N = 21$ is too small to deliver a decisive verdict on which family of macro predictors carries more weight, but the narrow inferential edge for the economic block is consistent with the individual-level finding. Chapter 8 closes the dissertation with a brief restatement of these contributions and their implications for the broader debate.